



ASME

ES 2026

20TH INTERNATIONAL CONFERENCE ON
ENERGY SUSTAINABILITY

HYATT REGENCY BELLEVUE

Practical Sizing and Control of Energy Storage for Hybrid Wind Farms

A Three-Layer Framework with Preliminary Wind-CAES Results

ASME ES 2026 · Session 05-02: Hybrid Energy Systems for Grid Integration or Microgrids II
Grid-Scale Hybrid Resource Planning and Storage Integration · ID 184121

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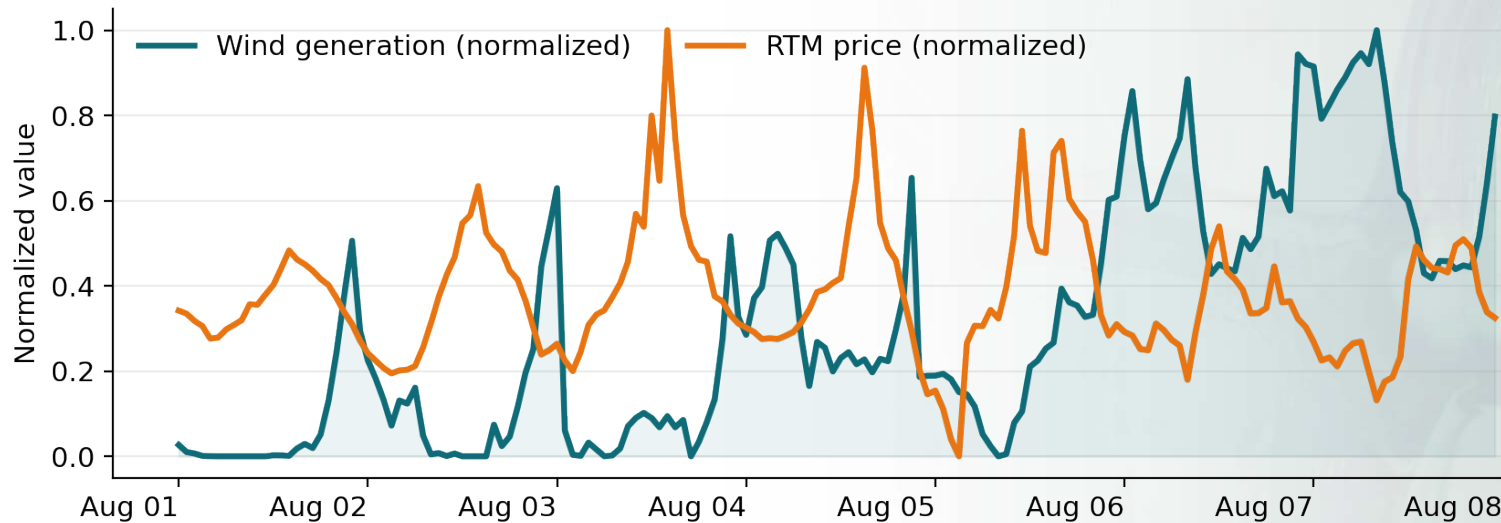
Why hybrid wind-storage value is hard to evaluate

Wind variability, price volatility, and forecast error interact with storage constraints.

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Wind output and market value often peak at different times



Storage is valuable only when the plant can shift energy from low-value or constrained hours to higher-value hours.

Perfect-foresight dispatch overstates the value available to a real controller.

Sizing choices and information quality should be evaluated together, not sequentially.

Revenue calculation

$$\text{Revenue} = \sum_t P_{\text{delivered},t} \cdot \text{price}_t \cdot \Delta t \quad | \quad \text{Revenue gain} = (R_{\text{case}} - R_{\text{baseline}}) / R_{\text{baseline}} \times 100\%$$

Submitted abstract scope → current presentation snapshot

The presentation remains aligned with the accepted abstract, but the numerical scope is CAES-centered and preliminary.

Accepted abstract scope	Presentation implementation
Offline optimal vs rolling-horizon operation	Main organizing comparison: perfect-information Oracle vs finite-information control
Storage sizing and control	Bounded power–duration sensitivity from preliminary A/B/C cases
Representative 250-MW wind plant	249 MW export cap and Pyron/Roscoe wind–market context
Multiple storage technologies	Framework-level discussion; numerical results focus on CAES-like parameters
DAM and RTM market data	RTM scoring in results; DAM retained as a limitation/future controlled information case

Technical Presentation Only

Preliminary results

CAES-centered

Research question

How much value survives when we move from perfect information to rolling-horizon operation?

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Can a forecast-aware controller retain the value predicted by idealized storage-sizing studies?

Information

What does the controller know about future wind and price?

Optimization

How do storage physics and market value shape planned dispatch?

Execution

How often is the plan updated when realized conditions arrive?

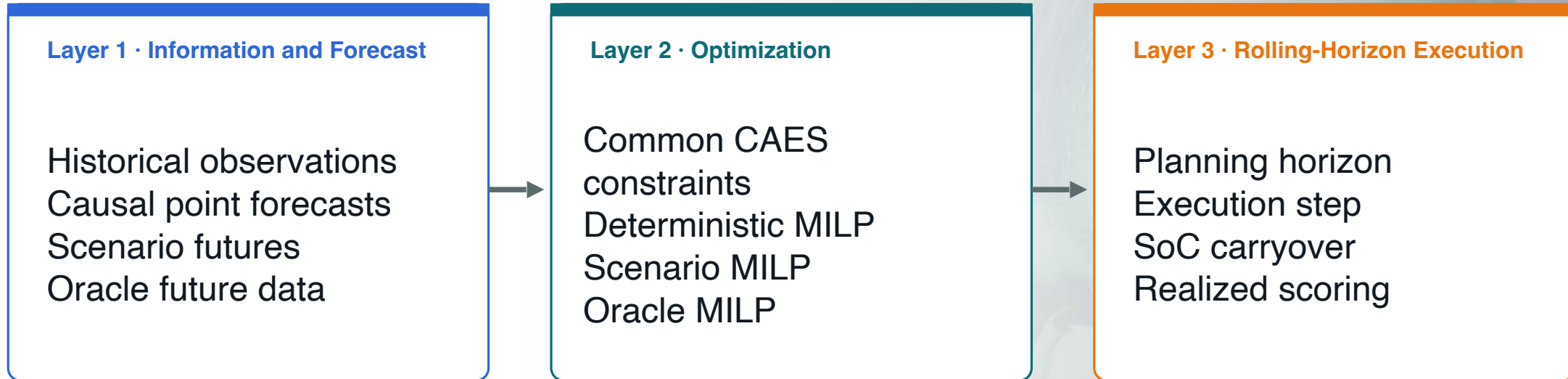
Key claim: perfect-information studies may not only overestimate value; they may also favor a different storage power–duration configuration than forecast-driven operation.

Three-layer framework

Separating information, optimization, and execution prevents apples-to-oranges comparisons.

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Case definitions and evaluation metrics cut across all three layers

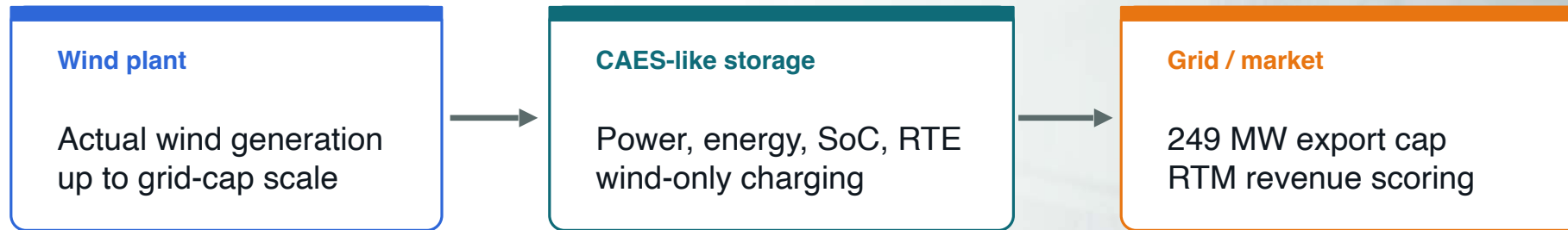
Wind-only · Constant-output baseload · Full/finite Oracle · Forecast-driven RH · Scenario-based RH

System definition and common physical constraints

Reference numerical system: approximately 250-MW wind plant + CAES-like storage.

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Parameter	Reference value
Wind plant / export scale	≈ 250 MW; 249 MW export cap
Reference CAES case	100 MW / 10 h / 1000 MWh
SoC limits	20–100% capacity
Round-trip efficiency	55%; discharge-side convention in current runs
Charging rule	Wind-only; no grid charging
Operation rule	No simultaneous charge and discharge; chronological SoC

Case definitions used throughout the talk

Each case differs by available information and control objective.

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Case	Information used	Purpose
Wind-Only Baseline	Current actual wind only	No-storage reference
100-MW Constant-Output Baseload Benchmark	Current wind + SoC; price not used for control	Rule-based stable-output benchmark
Full-Horizon Oracle MILP	Entire future wind + RTM price	Offline upper bound
H-Hour Oracle Rolling-Horizon MILP	Next H hours actual wind + RTM	Finite-look-ahead upper benchmark
Deterministic Forecast-Driven RH MILP	One causal future trajectory	Deployable point-forecast control
Scenario-Based RH MILP	Multiple possible futures with shared current action	Uncertainty-aware control

Terminology discipline: historical window = past; forecast/planning horizon = future; execution step = implemented portion of a plan.

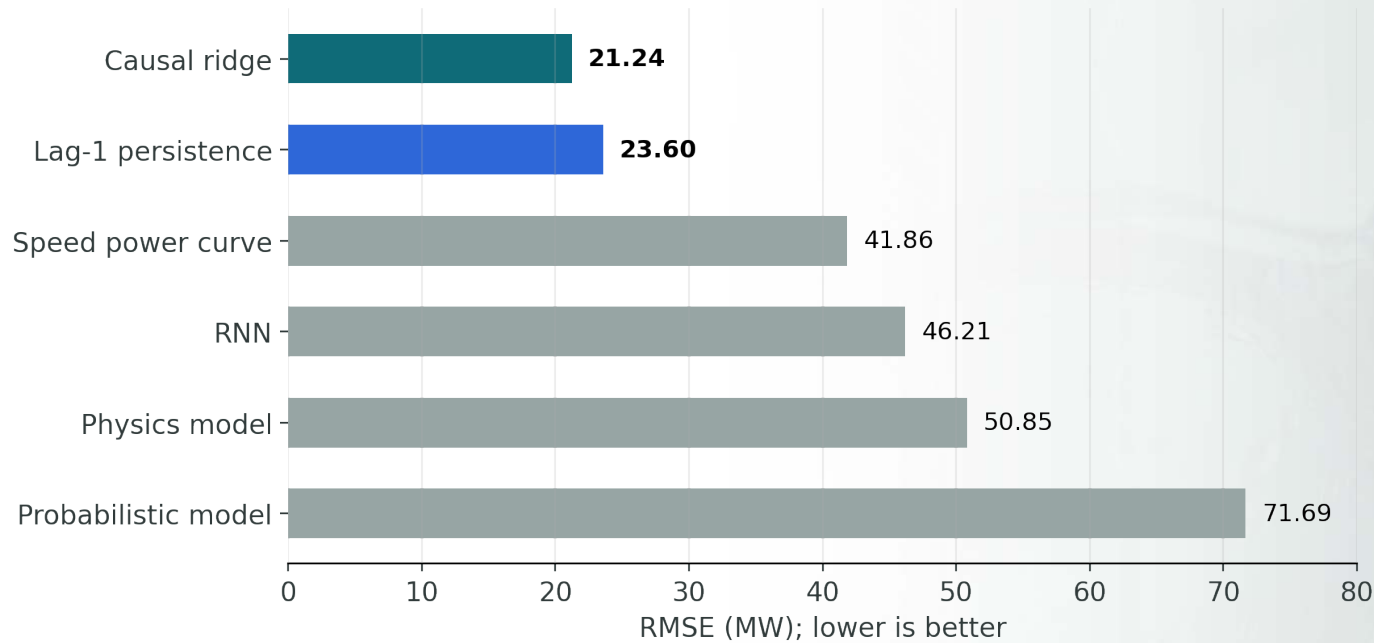
Layer 1 evidence: causal ridge provides a practical point forecast

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A simple model using recent behavior and wind-speed features outperformed saved alternatives in the current comparison.

Causal ridge was the strongest point forecast in the saved comparison



Inputs

Wind speed
Speed² and speed³
Recent power lags
Calendar features

Output

Predicted wind power trajectory
for dispatch planning

Interpretation

Better forecasts improve
the information layer, but
do not eliminate physical or
market limits.

Layer 2: storage-aware MILP formulation

The optimizer converts available information into planned direct delivery, charge, discharge, and SoC.

OBJECTIVE

Maximize realized market value subject to storage physics, wind availability, export limits, and operating mode constraints

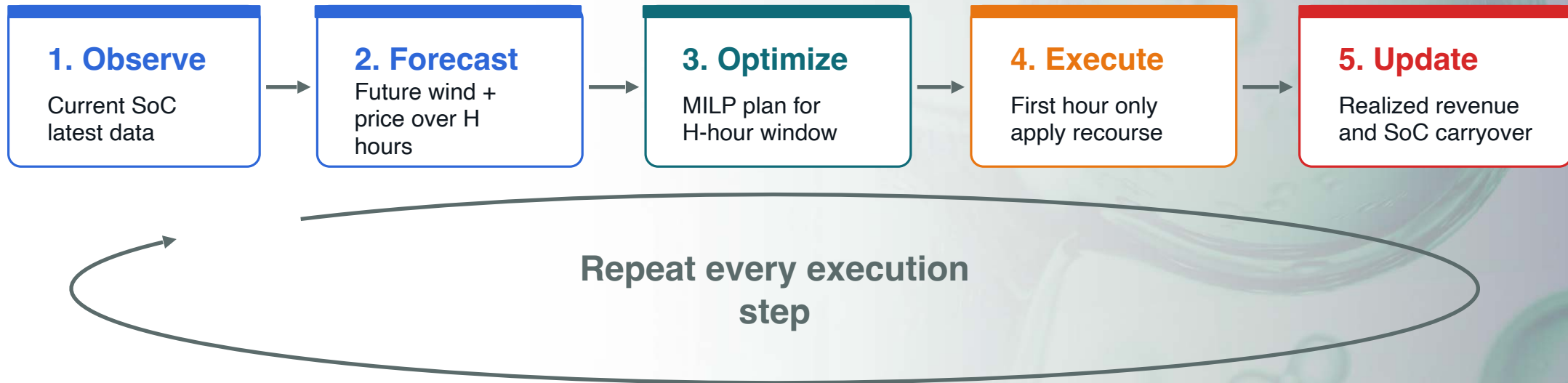
Constraint family	Meaning
SoC balance	Storage state carries forward hour by hour; discharge pays efficiency loss
SoC bounds	Storage cannot fall below the reserve or exceed capacity
Wind-only charging	Charging power cannot exceed available wind after direct delivery
No simultaneous modes	Binary variable prevents charging and discharging in the same hour
Grid export cap	Delivered power cannot exceed interconnection limit
Terminal rule	Selected per benchmark: full-period, annual, or report-only

Layer 3: rolling-horizon control and realized execution

Only executed actions create revenue; forecast-based plans must be converted into feasible realized operation.

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Critical distinction: planning horizon controls how far the optimization looks ahead; execution step controls how long the plan is actually implemented before replanning.

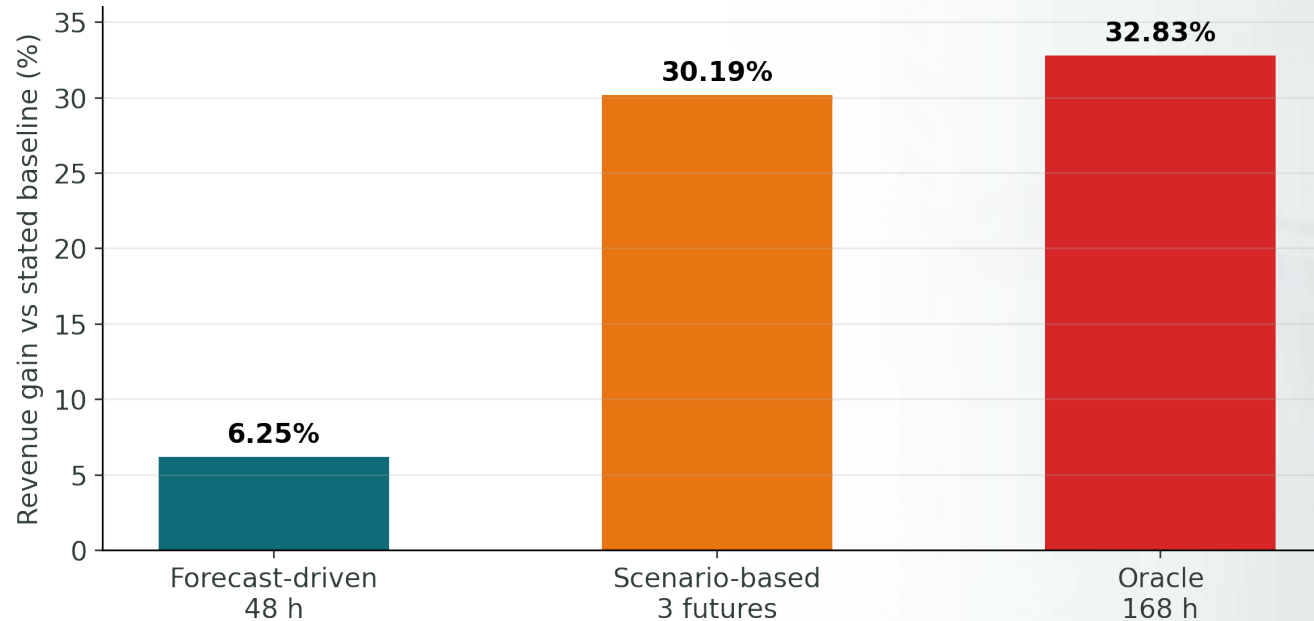
Preliminary ladder suggests value gains from information and uncertainty

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These values are useful for presentation, but they are not the final unified comparison table.

Preliminary ladder values: useful for storytelling, not yet a unified final comparison



Deterministic RH: 48 h was best in the current reported ladder.

Scenario RH: multiple forecast futures may add value beyond a point forecast.

Oracle: perfect future information remains the ceiling, not a deployable controller.

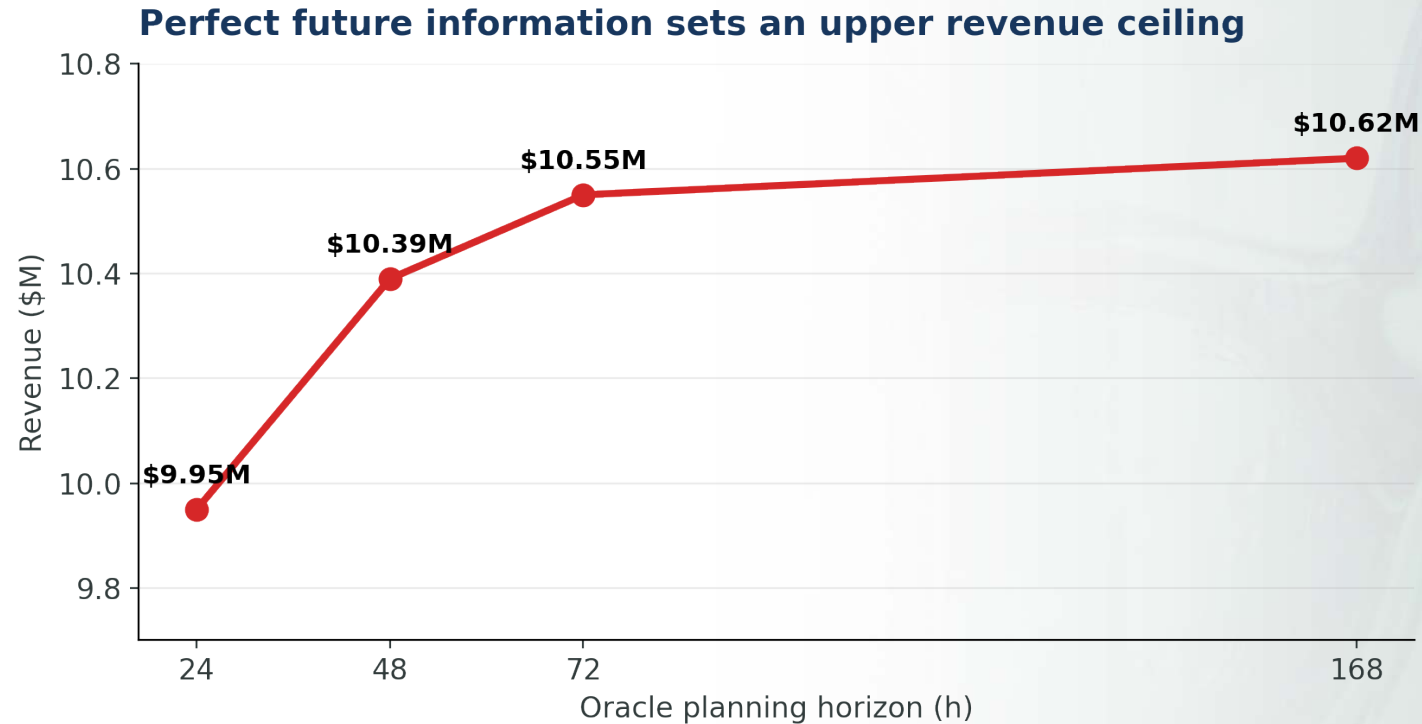
Use as preliminary evidence; avoid treating different ladder steps as a fully controlled comparison until storage configuration and baseline definitions are unified.

Oracle horizons quantify how perfect information changes the ceiling

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Finite-horizon Oracle cases show how much value is theoretically available when future wind and price are known.



Interpretation

Longer look-ahead gives the optimizer more opportunity to align storage actions with future wind and price patterns.

Important caveat

This is a perfect-future benchmark, not an implementable controller. It should be used to measure the gap faced by forecast-driven operation.

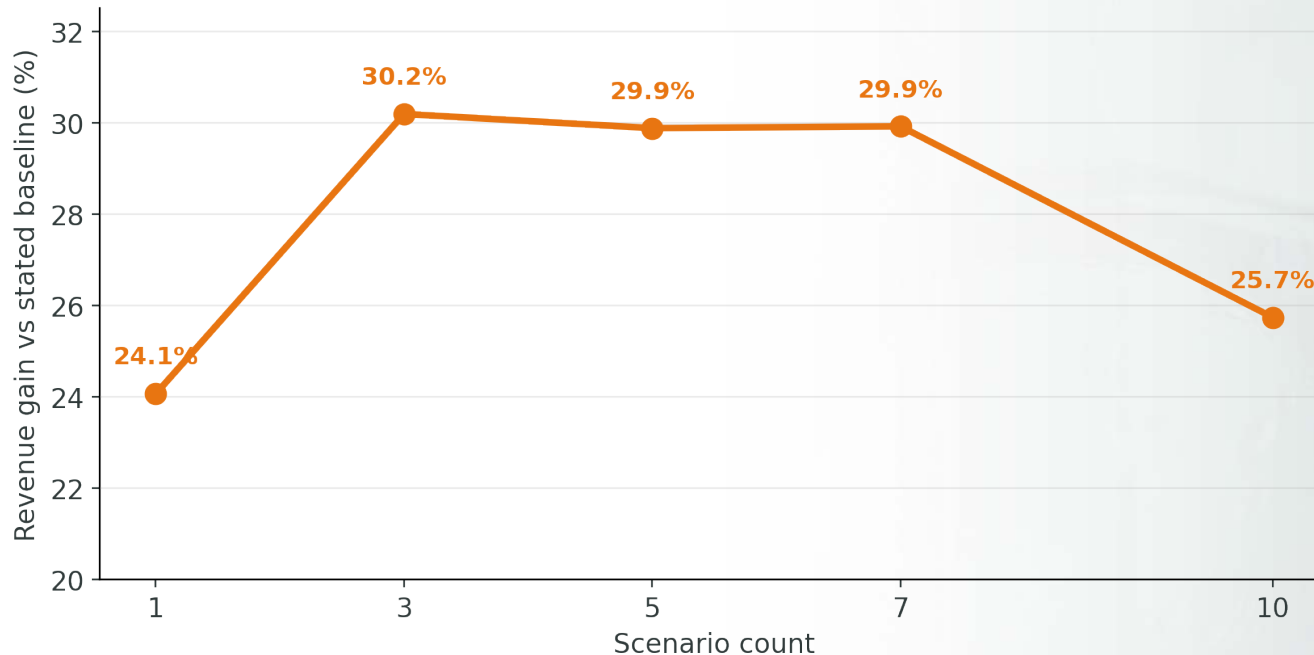
Scenario-based extension: beyond one point forecast

Multiple forecast futures can hedge decisions against forecast error, but scenario construction and QA remain ongoing work.

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Scenario-based dispatch may improve revenue, but the optimum count is not final



What it tests

Can a controller improve realized value by choosing actions that perform well across several possible futures?

What is not settled

The best scenario count is not yet a final design result; multi-seed and same-configuration checks are still needed.

Preliminary package provides a QA-backed configuration sensitivity snapshot

The current evidence compares three power–duration configurations under Oracle and causal operation.

Architecture	Power	Duration	Energy
A	100 MW	6 h	600 MWh
B	200 MW	3 h	600 MWh
C	100 MW	10 h	1000 MWh

Preliminary data scope

2020 only · 8,784 hours
Raw PYR_PYRON1 LMP
Negative prices retained
Wind-only charging

Preliminary hourly outputs passed the core constraint QA checks (zero violations)



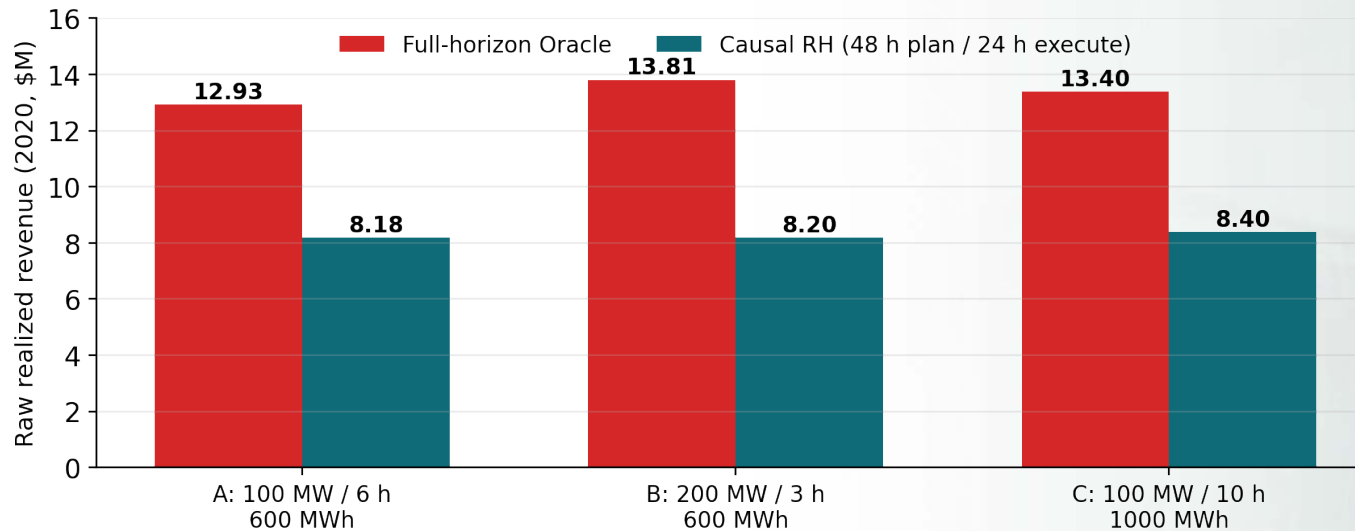
Configuration sensitivity: the ranking depends on information regime

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Raw realized revenue across three power–duration configurations changes when perfect information is replaced by causal operation.

Information regime changes the revenue ranking across storage configurations



Oracle ranking

$B > C > A$
High-power / short-duration is best with perfect information.

Causal ranking

$C > B > A$
Longer duration is best under the causal controller.

Not yet TEA

This is an operational-revenue ranking, not a cost-optimal sizing result.

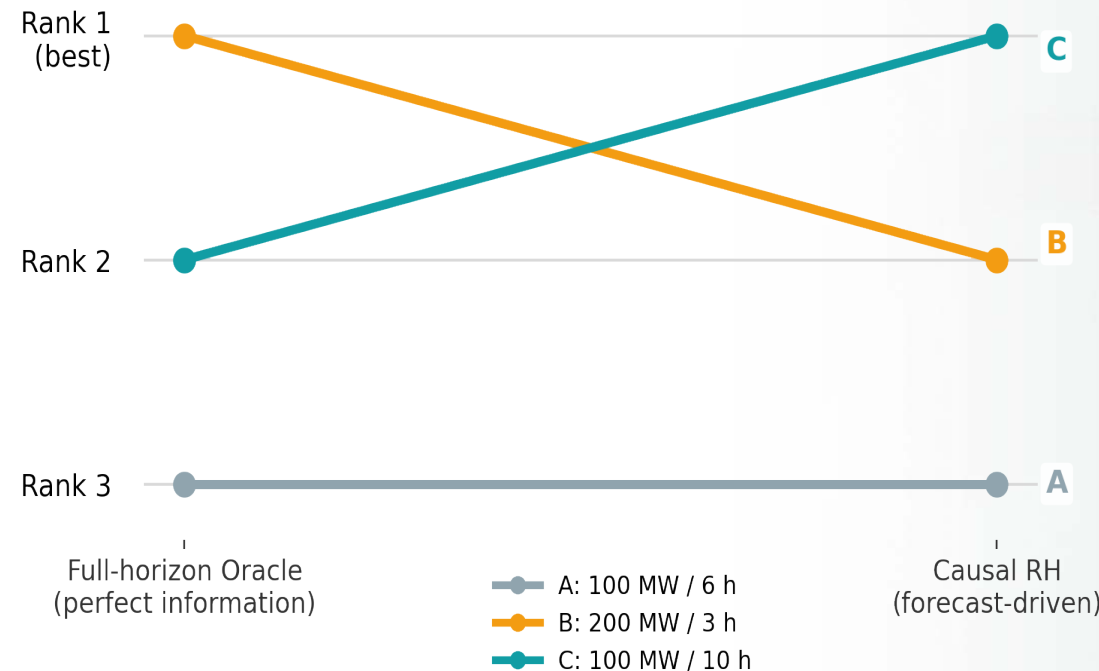
Engineering insight: perfect information can favor a different storage design

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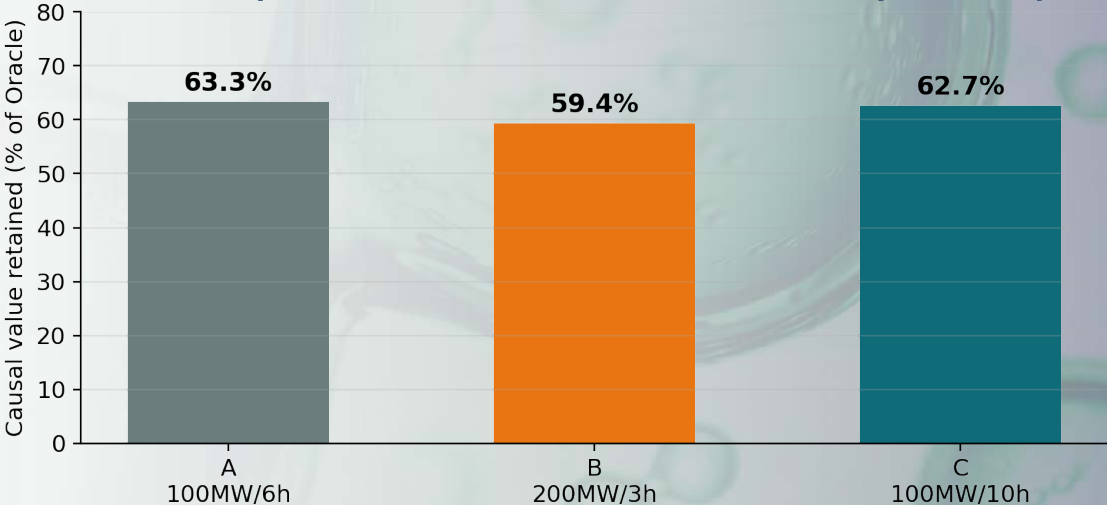
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The preferred power–duration choice can shift when forecast-driven operation replaces perfect foresight.

The preferred power-duration configuration flips when forecasts replace perfect information



How much perfect-information value is retained by causal operation?



Preliminary implication: sizing studies based only on perfect foresight may mislead both value estimation and design selection.

Scope boundaries and next steps

The framework is ready; several elements remain outside the current headline result.

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DAM price information

Tested but not yet cleanly matched to the PYR_PYRON1 RTM scoring signal; keep as a future controlled information case.

Newer proxy data

Useful for workflow sanity checks, but not official evidence until plant generation and market mappings are validated.

Scenario robustness

Promising preliminary result; needs same-configuration comparison, non-anticipativity audit, and repeated seeds or deterministic scenario design.

Techno-economics

Current preliminary result is revenue-only. Cost, O&M, cycling, and technology-specific constraints are next for sizing guidance.

Planned extension: unify the benchmark set under one configuration, then add DAM-informed and scenario-based controllers as controlled information upgrades.

Takeaways

Forecast-aware storage studies should couple information, optimization, and realized control.

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- 1. Perfect-information studies are useful as upper bounds, but not as deployable controllers.**
- 2. Rolling-horizon value depends jointly on forecast information, storage constraints, and recourse frequency.**
- 3. Preliminary evidence shows that information regime can change storage power–duration ranking.**

Acknowledgment: NSF REU student contributor **David Valenta**; prior REU contributions by **Zach Lawrence** and **Jessica Yao**; discussions with **Seyedehzeinab (Nora) Hosseiniimeni** on constrained storage dispatch.

Questions?